English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

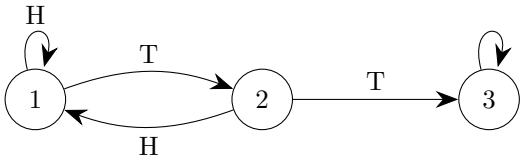
Applied Mathematics / Mathematics, 14 June 2026 Written examination

The exact grading formula will depend on the results. As an approximation, use 2 + the number of points. Every answer must be justified and supported by clear, detailed calculations. Electronic devices (except standard calculators) and additional materials are prohibited. Time allowed: 4 hours. Good luck!

Problem 1. (1.5 points)

1. (0.5 points) Let $Y \sim \text{Geo}(q)$[1] and let $X \mid Y = n \sim \text{NB}(n+1, q)$. Find the distribution, expectation and probability generating function of X.[2]

2. (0.5 points) Find the expected number of tosses of a coin with probability p of heads until two consecutive tails first occur. *Hint:* You may use the preceding parts.



3. (0.5 points) Find the expected number of occurrences of two consecutive tails in $n \geq 2$ tosses.

Problem 2. (1 point) Independently choose two points on the axes of a Cartesian coordinate system: A has a standard normal distribution on the horizontal axis and B on the vertical axis.[3]

1. (0.25 points) Find the distribution and expectation of $|A|$.

2. (0.75 points) Find the probability that (A, B) lies in the disc of radius 4 centred at $(0, 0)$. What is the probability that $\angle OAB \leq \pi/4$?

Problem 3. (1.25 points)

1. (0.75 points) Let $X_1, \ldots, X_n$ be iid with density $f(x; \theta) = 2x/\theta^2$ for $0 < x < \theta$, and 0 otherwise, where $\theta > 0$. Find the maximum likelihood estimator of θ and check whether it is unbiased and consistent.

2. (0.5 points) Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gamma}(\alpha, \beta)$, where $\alpha, \beta > 0$, with density $f(x) = \beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$ for $x > 0$, and 0 otherwise. Find method of moments estimators of α, β.

Problem 4. (1.25 points)

1. (0.25 points) A box contains 5 red, 4 blue and 3 green balls. Select 4 without replacement. Find the probability of selecting at least one of each colour.

2. (0.5 points) Let (X, Y) have joint density

$$f_{X,Y}(x, y) = \begin{cases} cxy, & 0 < x < 1,\ x^2 < y < \sqrt{x}, \\ 0, & \text{otherwise.} \end{cases}$$

   Find c, $\mathbb{E}[X]$, $\mathbb{E}[X \mid Y = 1/4]$ and $\mathbb{P}(X + Y < 1)$.

3. (0.25 points) Let $X_n \sim \text{Exp}(n)$ for $n \in \mathbb{N}$. Show that X_n converges in probability. Find the limiting distribution of nX_n.

4. (0.25 points) Let $\Theta \sim \text{Exp}(\lambda)$ and $X \mid \Theta = \theta \sim \text{Po}(\theta)$, with $\lambda > 0$. Find the conditional density of $\Theta \mid X = x$.

[1] Use the convention "number of failures before the first success".

[2] Recall $\sum_{j=0}^{\infty} \binom{m+j-1}{m-1} x^j = (1-x)^{-m}$ for $|x| < 1$ and $m \in \mathbb{N}$.

[3] Equivalently, use the points $(A, 0)$ and $(0, B)$, where $A, B \overset{iid}{\sim} N(0, 1)$.